

FARMER PARTICIPATORY CROP IMPROVEMENT. III. PARTICIPATORY PLANT BREEDING, A CASE STUDY FOR RICE IN NEPAL

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SUMMARY

A participatory plant breeding (PPB) programme was conducted for the high altitude areas of Nepal. The aim was to breed acceptable varieties with minimum use of resources and to utilize farmers' knowledge in the PPB programme. Farmer participation began at the F₅ stage and progress was followed over two seasons in two villages. Farmers proved to be willing participants and made selections in the segregating material, often with great success. Large differences in farmers' preferences between the F₅ bulks were found and the most preferred were adopted rapidly. The most adopted variety, Machhapuchhre-3, performed well in the formal trials system, and much better than the products from conventional centralized breeding. The PPB programme increased biodiversity in the two participating villages.

INTRODUCTION

In Nepal, as in many developing countries, rice must pass through a highly centralized process of varietal testing, release and certification, and typically this procedure takes many years. For example, 13 to 15 years are required for new cultivars in the National Rice Research Programme (NRRP) to reach farmers' fields (Sthapit, 1995).

Less than two varieties are released each year for farmers situated in a wide range of Nepalese rice ecosystems, and only 41 varieties have been released since 1966 (Sthapit, 1995). However, extreme agro-ecological diversity and ethnic-specific preferences of farmers demand the release of many location- and purpose-specific varieties. The limited varietal options available to the farmers lead to poor adoption levels. For example, only 10–11% of farming households were growing improved rice cultivars in a survey of more than 1688 households in 11 districts of Eastern and Western Nepal (LARC, 1995; Chemjong *et al.*, 1995).

Rice improvement programmes have been even less successful for the high-altitude areas (>1000 m) of Nepal where cold injury is common, partly because of the allocation of limited research resources by national and international programmes and partly due to a failure to use well-adapted material as parents. Most

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entries in cold-tolerant nurseries from the International Rice Research Institute (IRRI) failed to produce grain when grown in Nepal at Lumle (1450 m) and Chhomrong (2000 m) (Sthapit, 1992). Nepal has many genetically diverse, cold-tolerant rice landraces but little use has been made of them in the national breeding programmes.

Maurya *et al.* (1988), Galt (1989), Sperling *et al.* (1993), Sthapit *et al.* (1994), Joshi *et al.* (1995) and Witcombe *et al.* (1996) considered alternative approaches to address the problems of the limited varietal choice available to farmers. Traditionally, the development of rice varieties has been the task of breeders, and farmers are merely the recipients of finished products. The objective of the programme was to involve farmers at a much earlier stage in order to produce farmer-acceptable cultivars more efficiently. Participatory plant breeding (PPB) was employed for high-altitude rice breeding using crosses involving locally-adapted parental material. It was intended that PPB would minimize resource use, produce cultivars that were acceptable to farmers, and increase the genetic diversity of rice in the participating villages.

MATERIALS AND METHODS

Overall breeding scheme

The PPB programme involved farmers selecting from F₅ bulks onwards (Fig. 1). To satisfy formal release requirements there was a 'parallel' breeder-led scheme (Witcombe *et al.*, 1996) in which the best material was entered into the formal trials system and purified by progeny row selection for certified seed production (Fig. 1).

Identification of sites

Cold air and water at Chhomrong (2000 m) caused a high level of spikelet sterility. There was also a naturally high disease pressure for bacterial sheath brown rot disease caused by *Pseudomonas fuscovaginae* (Sthapit *et al.*, 1995) and blast caused by *Pyricularia oryzae* (Sthapit, 1995). Environmentally-induced spikelet sterility was lower at Ghandruk (2000 m), disease pressure was less and socio-economic conditions were favourable. Rapport had been established between scientists and farmers over many years of research by the Lumle Agricultural Research Centre (LARC) in these villages. It was known from participatory rural appraisals that farmers wished to improve the milled grain colour and the quality of their local rice. Hence, the goal of the programme was to develop a rice variety with white grain, a trait preferred by the local community. 'White grain' was used as a catch phrase to draw the farmers' interest.

Identification of segregating materials

Several diverse crosses were used (Table 1), and most involved the rice cultivar Chhomrong Dhan (Ch. Dhan), a pure line with resistance to sheath brown rot selected at the Lumle Agricultural Research Centre from a cold-tolerant

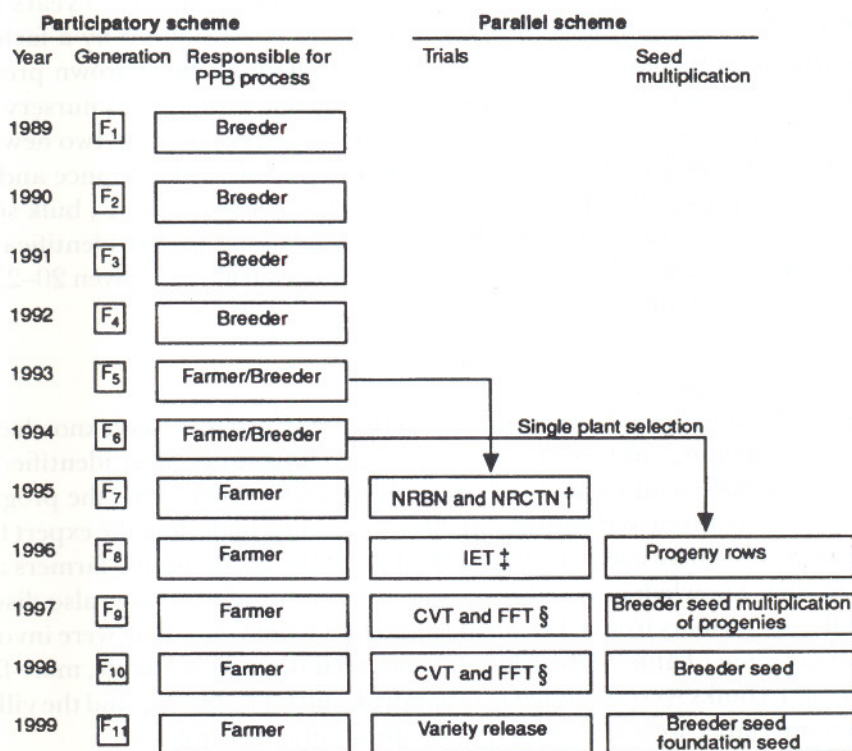


Fig. 1. Participatory plant breeding (PPB) using a parallel breeding scheme for the formal release system.

†NRBN = National Rice Blast Nursery, NRCTN = National Rice Cold Tolerance Nursery; ‡IET = Initial Evaluation Trial; §CVT = Coordinated Varietal Trial, FFT = Farmers Field Trial.

Table 1. The five parents of the four crosses† from which the F₅ bulks were derived

Genotype	Sub-species	Origin	Source	Important traits
Fuji-102	<i>Japonica</i>	Japan	IRCTN†	Cold tolerant, dwarf, white grain and good quality; not released in Nepal
Chhomrong Dhan (Ch. Dhan)	<i>Japonica</i>	Nepal	Local selection	Multiple resistance to cold, sheath-brown rot and blast diseases, medium plant height, red grain; released in Nepal
K-332	?	India	IRCTN	Cold tolerant, released in Kashmir, blast resistant
NR-10157-2B-2	<i>Indica</i>	Nepal	Breeding line	Early, cold tolerant, fine quality, not released in Nepal but sister line adopted by farmers
Stejaree-45	<i>Japonica</i>	Russia	IRCTN	Cold tolerant, coarse grain

†Shown in the first column of Table 2; ‡IRCTN = International Rice Cold Tolerance Nursery (International Rice Research Institute).

rot. It was released in 1991 for the high-hill areas of Nepal and is the only released cultivar that is well adapted to altitudes above 1500 m. About 25 years ago this landrace, from an unknown location in India, was introduced by a farmer into Ghandruk and Chhomrong villages where rice had not been grown previously. Six lines, from two crosses, were selected by breeders from the F_4 nursery in 1992 and four lines were selected from one of the previous crosses and two new crosses in 1993. Until the F_4 , breeders selected these lines for cold tolerance and sheath brown rot resistance by the pedigree-bulk method of selection. F_5 bulk seed was harvested from each F_4 row and given a local name to aid identification by farmers. In 1993 and 1994, farmers identified previously were given 20–25 g seed of the F_5 bulks (Table 2).

Identification of expert farmers

Expert farmers from Chhomrong and Ghandruk villages, with knowledge and skill in rice farming, and willing to participate in this study, were identified by the local community with the help of researchers. In October 1992, the programme objectives were discussed with about 20 rice growers, including the expert farmers identified previously in each village. The role of the participating farmers and the level of their knowledge on selection and inheritance of traits was also discussed. Initially, six farmers from Chhomrong, and four from Ghandruk were involved in the programme (Table 2). In 1994, as more seed became available, more farmers and more F_5 bulks were included in Ghandruk and Chhomrong, and the villages of Lumle and Sabet were also added to the programme (Table 2).

Both males and females from the same household were included in the study to take account of gender-specific knowledge and perceptions. Women were always included in the post-harvest evaluations, whereas men were more involved in the evaluation of pre-harvest and yield traits.

Management of the trials

The farmers were asked to grow and manage the bulks according to their normal practices, except that the test entries had to be kept separate in the field and store. Farmers assessed and selected according to their own criteria, and the methods they used were monitored and recorded by field staff. In each year, sowing in nurseries and transplanting of all the test varieties and the local control (Ch. Dhan) were done within a week. At each plot, sign-boards were erected to display the name of the variety and the farmer so that the attention of non-participating farmers was drawn.

Training in plant selection

Farmers were informed that the offspring from two contrasting parent plants would vary (segregate) for traits. They were told that selection within variable traits should be done for two to three years in the F_5 bulk seed that they had been given, until the trait varied no longer. They were also told that some traits, such as

Table 2. *Farmers and genetic materials in the participatory breeding programme, 1993 and 1994*

Parentage	Vernacular name of F ₅ bulk	Name of the participating household member			
		Chhomrong	Year†	Ghandruk and Lumle	Year
Fuji 102 × Ch. Dhan‡	Machhapuchhre-1 (M-1)	R. B. Gurung	1993	M. K. Gurung	1993
Fuji 102 × Ch. Dhan	Machhapuchhre-2 (M-2)	K. B. Gurung	1993, 1994	J. B. Gurung	1993, 1994
Fuji 102 × Ch. Dhan	Machhapuchhre-3 (M-3)	J. B. Gurung	1993, 1994	R. B. Gurung	1993, 1994
Fuji 102 × Ch. Dhan	Machhapuchhre-4 (M-4)	C. B. Gurung	1993, 1994	P. B. Gurung	1993, 1994
Fuji 102 × Ch. Dhan	Machhapuchhre-5 (M-5)	R. K. Gurung	1993	K. M. Gurung	1993
K332 × NR10157-2B-2	Himchuli-1 (H-1)	M. B. Gurung	1993	B. B. Paudel§	1993
K332 × NR10157-2B-2	Himchuli-2 (H-2)	H. B. Gurung	1994	D. J. Devkota§	1994
Stejaree 45 × Ch. Dhan	Nilgiri-1 (N-1)	C. B. Gurung	1994	J. N. Devkota§	1994
Fuji 102 × Ch. Dhan	Machhapuchhre-6 (M-6)	J. B. Gurung	1994	—	
Fuji 102 × Ch. Dhan	Machhapuchhre-7 (M-7)	I. B. Gurung	1994	—	

†Year of first participation; ‡Ch. Dhan = Chhomrong Dhan; §At Lumle.

grain colour, were 'strongly' inherited and therefore good targets for selection. This training in the simple principles of genetics and heritability was given to complement farmers' indigenous knowledge. After each harvest, farmers were asked to return half of the selected seed to the Lumle Agricultural Research Centre for on-station testing. The retained seed was used by the farmers in any way they chose.

Farmers' preference ranking

In 1993, farmers' plots in each village were visited during 'farm walks' by the participating farmers and the breeders, but no preference ranking was done. In 1994 and 1995, the on-farm and experimental plots at Chhomrong and Ghandruk villages were monitored jointly by participating and non-participating farmers, breeders, a socio-economist and affiliated researchers. At the outset, a breeder explained the purpose of the farm walk, and everyone visited all the test plots in each village.

In 1994, after the farm walk, farmers were asked individually to rank varieties on a scale from 1 (excellent) to 7 (worst). Farmers were helped by researchers to list positive and negative characteristics for each variety using an open format questionnaire. The extent of agreement of the preference rankings of individuals among male, among female and among all farmers was assessed using the Kendall coefficient of concordance W (Siegel, 1956), a measure of the agreement among a group of judges. The extent of agreement between two groups of judges, such as breeders and farmers, was measured using the overall orders of each group and Spearman's rank order correlation coefficient r_s (Steel and Torrie, 1960).

On-farm yield measurement

In 1994, the rice crops of all participant farmers were sampled by harvesting a 1 m² plot. After threshing, grain yield was measured and adjusted to 12% moisture content. Plant height and fresh straw yield at harvest were also recorded.

On-station yield trials

Machhapuchhre-3 (M-3), selected by R. B. Gurung from Ghandruk-6, was the most popular F_5 bulk in the 1993 farm walks, even though no formal preference ranking was made. In 1994, M-3 was entered in the first stage of national varietal testing, the National Rice Cold Tolerance Nursery (NRCTN). This nursery was located at Chhomrong, Shera (1250 m), and at the Kavre Agricultural Farm (1700 m) during the 1994 rainy season. Trials were designed in a randomized complete block design with two replicates at Shera and Chhomrong. At Kavre the trial had only a single replicate.

The trials were managed by researchers using standard recommended agronomic practices. The seedlings were transplanted from nursery beds into fields in mid-July 1994 and were given 60–30–20 kg NPK ha⁻¹. Plot size varied from 1.2 to 4 m². Data on agronomic traits and disease incidence related to cold tolerance

were measured. Analyses of variance were used in the replicated trials to evaluate the significance of differences between the varieties.

Post-harvest evaluations by women farmers

A post-harvest evaluation was carried out three months after the harvest. In the 1994 season, this study involved only those varieties that farmers intended to grow again in the next season. The questionnaire used in the evaluation was designed after discussions with women farmers. Farmers consumed the rice after using local methods of milling and cooking. Seven household level questionnaires on post-harvest evaluation were completed.

Monitoring of varietal spread

Varietal spread was monitored with all participant farmers in 1993, 1994 and 1995. The reasons for the adoption or rejection of varieties were also recorded.

RESULTS

Farmers' selection of trial sites and performance of the F_5 bulks

Most of the farmers conducted their trial in conditions of medium fertility, at altitudes from 1400 to 2000 m. Four examples illustrate how farmers often carried out risk-aversion strategies. H. B. Gurung of Lumle, transplanted Himchuli-2 in a small area where water was at its coldest on first entering from the stream, and where, as a result, severe spikelet sterility had always been present in his local rice. D. Devkota of Lumle evaluated Himchuli-2 under highly fertile conditions that caused the best local varieties to be partially sterile. Himchuli-2 yielded well, and a greater area was planted in the following year. The most extreme risk-averse strategy was adopted by J. N. Devkota of Lumle who transplanted Nilgiri-1 into a plot where he had never seen a rice plant produce a single grain. The plot had an inlet of cold water from a mountain stream carrying animal manure. Cold water in the presence of high fertility induces high spikelet sterility (Sthapit, 1992), but the variety yielded 3.3 t ha^{-1} and the farmer was excited by this unexpected result. He said he would grow Nilgiri-1 in better land in larger areas, but later changed his mind because the variety exhibited poor post-harvest traits. R. B. Gurung of Ghandruk first tried M-3 in the worst parts of his land in a corner of a high terrace that received little sunlight and cold irrigation water. He selected the best plants from this area, and increased the land he devoted to the variety from 6 to 1250 m^2 within two years.

Farmers' perceptions of the F_5 bulks

In 1993, most of the entries included in the participatory programme were cold tolerant. However, the performance of Himchuli-1 was not very satisfactory due to high panicle sterility. Among the Machhapuchhre (M) lines, farmers dropped M-1 after the first year because it was very dwarf and was more sterile than Ch. Dhan, and M-5 was dropped because of its low yield. Although the performance

Table 3. *Preference ranking† of rice varieties when near to maturity under farmer-managed conditions, Chhomrong, 1994*

Variety	Preference ranking				
	Male‡ farmers (n = 9)¶	Female‡ farmers (n = 5)	All farmers§ (n = 14)	Breeders§ (n = 2)	Overall (n = 16)
Machhapuchhre-2	2	2	2	2	2
Machhapuchhre-3	1	1	1	1	1
Machhapuchhre-4	3	3	3	3	3
Machhapuchhre-6	4	6	4	3	4
Machhapuchhre-7	6	4	5	4	5
Himchuli-2	7	7	7	5	7
Nilgiri-1	5	5	6	4	6
Kendall's W	0.81	0.83	0.67	0.96	0.77

†Preference ranking on a scale from 1 (excellent) to 7 (worst); ‡Rank correlation (r_s) between male and female farmers, $r_s = 0.82$; §Rank correlation (r_s) between breeders and farmers, $r_s = 0.82$; ¶n = number in group; ||Kendall's 'W' is measured on the scale 0 (no agreement) to 1 (perfect agreement).

of M-2 was average in 1993, farmers decided to evaluate it again in the next season. M-3 and M-4 appeared promising and farmers retained their seed for a further season of testing.

In Chhomrong, in October 1994, 14 farmers from 10 farming households took part in the farm walk. The comparative performance of the three varieties retained from 1993, and the four additional varieties given in 1994 (Table 2) was judged by preference ranking. M-3 had the highest overall ranking followed by M-2 and M-4. Himchuli-2 was ranked lowest (Table 3) because the panicles were small and had poor seed set.

There was significant agreement between the evaluation scores of men and women farmers in Chhomrong village (Table 3). Agreement between the two breeders was higher than amongst the 14 farmers. However, the higher consistency amongst the breeders' judgements is almost certainly because it is easier for two people to have common views than fourteen. Agreement between breeders and farmers was high ($r_s = 0.82$). This is not unexpected because the farmers were carefully chosen for their skills, and the breeders were aware of farmers' preferences.

Of the 14 participating farmers, two at Ghandruk and three at Chhomrong, retained seed for further testing and evaluation (Table 2).

Yield from farmer-managed plots

Grain yields and other agronomic traits were measured for a 1-m² sample of the on-farm plots used for the preference ranking (Table 4). Although single crop harvest data cannot be analysed statistically, there was good agreement between farmers' perceptions of the variety and crop harvest results. The preferred variety, M-3, yielded the same or more grain than Ch. Dhan while the other test varieties yielded less than Ch. Dhan (Table 4). The plant height and straw yield of M-3 was also superior to Ch. Dhan.

Table 4. Grain yield and other agronomic traits of rice harvested from a 1 m² plot in farmers' fields at Chhomrong and Ghandruk (1700–2000 m), 1994

Variety	Grain yield (t ha ⁻¹)		Plant height (cm)		Straw yield (t ha ⁻¹) at harvest		Tillers plant ⁻¹	
	Chhomrong	Ghandruk	Chhomrong	Ghandruk	Chhomrong	Ghandruk	Chhomrong	Ghandruk
Machhapuchhre-2	5.22 (6.35) [†]	—	116 (118)	—	20.0 (18)	—	8.0 (5)	—
Machhapuchhre-3	6.32 (5.06)	6.50 (6.50)	123 (110)	130 (119)	24.0 (17)	29.0 (24.0)	6.7 (4)	7.0 (2.5)
Machhapuchhre-4	3.55 (5.06)	3.35 (3.03)	114 (116)	118 (120)	11.2 (25)	9.2 (10.0)	5.6 (5)	6.2 (3.0)
Machhapuchhre-6	3.14 (4.09)	—	100 (116)	—	5.0 (5)	—	5.2 (6)	—
Machhapuchhre-7	3.23 (4.72)	—	111 (120)	—	16.0 (11)	—	6.0 (4)	—
Himchuli-2	1.43 (5.06)	—	117 (125)	—	20.0 (26)	—	4.0 (5)	—
Nilgiri-1	3.52 (5.06)	—	110 (106)	—	18.0 (18)	—	5.6 (5)	—
Mean	3.77 (5.05)		113 (116)		16.3 (17)		5.8 (5)	

[†]Figures in parentheses are data for the local control variety, Chhomrong Dhan grown adjacent to a test variety, n = 1.

Farmers' methods of plant selection

Farmers used various criteria to select the best plants. At maturity some farmers selected for grain colour without considering plant height or how early the plants matured. This resulted in a segregating crop that was unimpressive in the following year. Other farmers bulked seed only from plants of similar height, maturity and grain colour. Apart from selecting for grain colour, farmers selected plants with long, compact and drooping panicles, good grain setting, and freedom from sheath brown rot disease.

For example, M. B. Gurung selected plants of M-3 with intermediate height, compact panicles and filled grains. He selected plants where the flag leaves remained green after maturity, because he thought that the straw quality of such plants would be better. He also considered the maturity of test plants compared with the local variety grown alongside. P. B. Gurung, a retired livestock field officer at the Lumle Agricultural Research Centre, had some knowledge of selection and heritability. First he removed undesirable plants that were dwarf, too tall, or diseased and then he selected panicles from white-grained plants with high grain set and long panicles. Another farmer, R. B. Gurung, who tested M-3 in the worst plots of his land, harvested the best tall plants. He examined the grain colour of each selected plant and bulked the seed from those plants with white-grained panicles. He planted this seed in 1994 in his best field.

The major selection criteria were grain colour, yield, plant height, and maturity relative to the local variety. Besides grain yield, farmers considered how densely grain had set in the panicle, panicle length and tillering ability. Single-plant selection was usually undertaken only in the best plots, because farmers rarely bothered to select the few best plants from an entry when its overall appearance was poor. Farmers were willing to take time and care in selection in the first two years, but the amount of care and time increased in the third year of the programme when they appreciated the potential value of the exercise more fully.

In the villages of Chhomrong and Ghandruk farmers from similar ethnic groups and farming backgrounds selected from the same F₅ bulk seed. However, the end results of selection on the same material varied. The resultant bulks, M-3C from Chhomrong and M-3G from Ghandruk, were very different when grown alongside each other at Lumle.

Post-harvest assessment by women farmers

Women farmers said they would like to decide which were the best rice varieties after post-harvest evaluation and they used several qualitative criteria to assess quality (Table 5). Varieties M-6 and M-7 had a poor score for several traits, whereas M-2, M-3 and M-4 were liked for most traits, and farmers expanded their area of cultivation. White-grained rice was preferable to red-pericarped rice as it saved women time in milling (Table 6, comment 2). The post-harvest evaluation was the ultimate criterion upon which farmers either rejected or adopted the varieties. For example, J. N. Devkota was very impressed with the performance of Nilgiri-1 but decided not to grow it outside his problem plots because of its poor

Table 5. *Post-harvest assessment† of various rice varieties relative to the local control variety Chhomrong Dhan, Chhomrong, 1994*

Traits‡	M-2	M-3	M-4	M-6	M-7
Milling %	50.0	45.0	56.0	45.0	40.0
Broken rice %	1.9	5.0	1.3	5.0	5.0
Water-absorbance capacity	+	+	+	—	+
Elongation capacity	+	=	+	—	+
Aroma	+	—	—	—§	—
Dryness	+	+	+	—	+
Stickiness	+	+	+	—	+
Taste	+	+	+	+	+
Appetite delay	+	+	+	—	+

† + Better than local control, = Equivalent to local control; — Inferior to local control; ‡ Assessment was by farmers after cooking the rice by their own method and comparing with Chhomrong Dhan; § Farmers disliked the peculiar aroma of M-6.

taste. M-6, which performed reasonably well in the field, was rejected because of the peculiar smell of the cooked rice.

The milling percentage of all rice varieties was poor (40–56% by volume, 1.3–5% broken rice), but farmers considered 50% milling recovery acceptable. M-4 had the highest milling recovery (56%) and the lowest percentage of broken rice. It was easier and quicker to mill by *dhiki* (the local implement for dehushing which is usually operated by two women) because it has very thin husk. P. B. Gurung was so impressed with M-4 that he sowed all the seed he had, both selected and unselected. Farmers were reluctant to continue with M-7 because its milling recovery was poor (40%). The milling recovery of the other cultivars did not differ significantly from the local variety.

The best panicles selected by farmers were kept for seed, whereas the grain from the late tillers and small panicles were used for the post-harvest evaluation. This resulted in a lower milling percentage and a high proportion of broken rice. Farmers considered that a variety with a similar milling percentage to the local control variety would have had a better milling recovery if the best panicles had been evaluated.

Monitoring of spread

The preferred varieties had spread substantially within two years of their introduction (Figure 2). M. B. Gurung from Chhomrong, who obtained half the seed of M-3 from his friend J. B. Gurung, selected from M-3 successfully (his selection is identified as M-3C), and increased its area from 3 to 150 m² within a year. Similarly, R. B. Gurung from Ghandruk also selected M-3 (identified as M-3G) and increased its area from 6 m² in 1993 to 50 m² in 1994. In 1995, he sowed sufficient for a cropped area of about 1250 m² (Fig. 2). This variety occupied 3% of his rice fields after the first year of selection and over 60% after the second. The

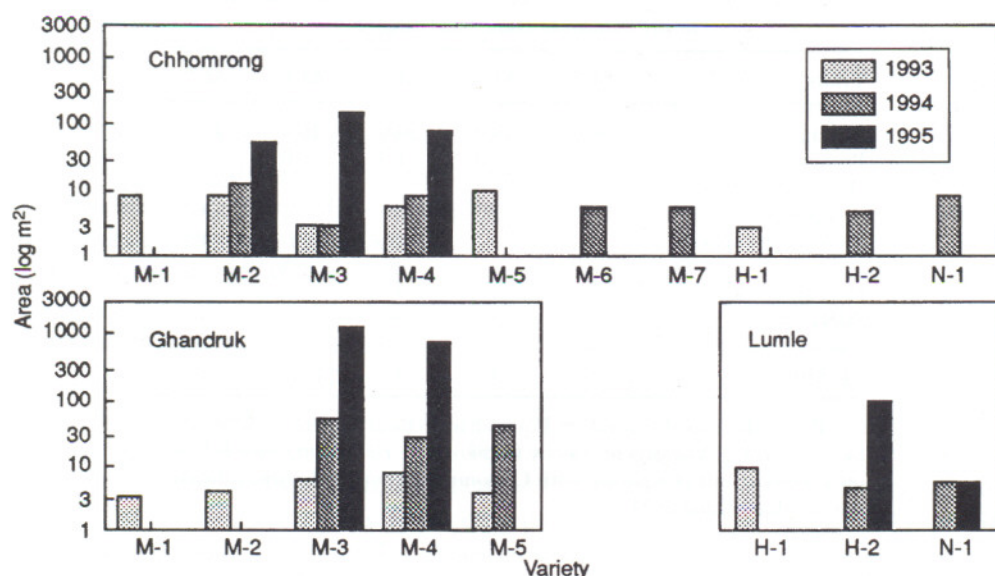


Fig. 2. The area of adoption of new rice bulks in Chhomrong and Ghandruk villages, 1993–95. M-1 to M-7 = Machhapuchhre-1 to Machhapuchhre-7, H-1 and H-2 = Himchuli-1 and Himchuli-2, N-1 = Nilgiri-1.

most preferred bulks spread at a logarithmic rate, whilst other bulks were abandoned after one or two years (Fig. 2). Whether the bulks were grown again, or not, was also location dependent.

Farmers' perception of the programme

At the beginning of the programme, farmers were not very enthusiastic. They were accustomed to conventional, labour-intensive, on-farm trials managed by researchers. However, farmers' participation increased as the programme began to produce encouraging results. Comments made during field visits reflected the satisfaction of the participating farmers (Table 6). P. Shrestha, a member of staff at the Lumle Agricultural Research Centre, reported that: 'In the beginning it was difficult to find a farmer willing to participate but now the whole community is willing to offer their help.'

On-station yield trial

In on-station yield trials, the performance of farmers' selected varieties was compared with varieties bred through centralized breeding programmes. Results from these trials could be used to satisfy the variety release regulations by validating participatory on-farm data with on-station data. At Chhomrong, M-3 performed as well as the local control variety Ch. Dhan ($p \leq 0.05$) (Table 7). Both varieties showed superior resistance to sheath brown rot disease (Table 7) and cold temperatures at anthesis. At Kavre, the farmer-selected M-3 yielded 6.7 t

Table 6. Farmers' perceptions of the participatory breeding programme at Chhomrong and Ghandruk, 1994

Location	Date	Gender	Size of group	Comments
Chhomrong	7 Oct 94	Male	9	'Any rice variety that grows at this altitude is good. We need a variety which yields more and gives more straw. If variety has white grain colour it is a bonus. We will further select the plants to grow in larger plots next year.'
Chhomrong	7 Oct 94	Female	5	'If we can change our local rice into white grain rice it will save a lot of our (womens') time. We spend one to two hours extra time to dehusk rice until we get white grain.' 'Machhapuchhre-3 has both more grain and more straw. It has long panicles and grains are plenty. It matures at the same time as the local variety and the plants are taller. If it tastes good we would like to continue with this variety.'
Ghandruk	9 Oct 94	Male	1	'In the beginning I was not interested to involve myself but when LARC scientists told me that it has white grain then I became curious. I first tried it in the worst parts of my land. I saw tall plants producing really good panicles. I selected all of the best plants with white grain and maturity similar to our local variety. This year it looks really good and better than last year. Now I am happy to grow it in all my plots. I have no plan to share the seeds until I fulfill my requirement.'

grain ha⁻¹ compared with 5.4 t for the local control, Ch. Dhan. It also out-yielded other superior local varieties known for their high cold tolerance in Nepal (Table 7). M-3 also showed a comparable incidence of sheath brown rot and neck blast under natural inoculum pressure when compared with most breeder-selected varieties. One of the farmer-selected rice varieties M-3G (Machhapuchhre-3 Ghandruk), which was multiplied and included in 1994 in the national varietal testing system of the Nepal Agricultural Research Council (1994), was promoted to the Initial Evaluation Trial as a promising entry.

DISCUSSION

Many workers have advocated decentralization of research for plant breeding (Maurya *et al.*, 1988; Farrington and Martin, 1988; Galt, 1989; Joshi and Sthapit, 1990; Sperling *et al.*, 1993; Sthapit *et al.*, 1994; Joshi and Witcombe, 1996; Witcombe *et al.*, 1996). Formal research systems in developing countries are highly centralized and do not target the problems of resource-poor farmers, as evidenced by the poor adoption of officially released rice varieties in India (Maurya *et al.*, 1988; Joshi and Witcombe, 1996) and Nepal (LARC, 1995; Chemjong *et al.*, 1995).

We believe that this is the first well-described example of participatory plant breeding (PPB) in which farmers have selected within segregating lines. It has

Table 7. Performance of *Machhapuchhre-3 (M-3)* in comparison with the best yielding entries from conventional breeding and the local control varieties in the National Rice Cold Tolerance Nursery (NRCTN), summer 1994 at Shera (1250 m), Kavre (1700 m) and Chhomrong (2000 m) under researcher-managed conditions

Variety	Grain yield (t ha ⁻¹)			Time to 50% heading (days)			Plant height (cm)			ShBR†		
	1250 m	1700 m	2000 m	1250 m	1700 m	2000 m	1250 m	1700 m	2000 m	1250 m	1700 m	2000 m
M-3 (W)‡	5.2	6.7	4.3	96	118	156	120	136	119	11	10	2
Ch. Dhan (R)‡	6.1	5.4	4.9	91	105	147	130	116	127	4	12	5
Darmali (W)	1.9	5.0	0	111	123	180	137	116	129	5	4	2
J. Marshi (W)	1.6	6.0	2.7	90	145	131	119	109	141	13	36	12
Best (n=5)§	5.8±0.38	5.4±1.1	2.4±0.5	99±2.1	117±4.8	151±5.7	133±4.0	100±3.3	124±5.9	5.8±2.2	15.2±3.6	8.5±2.5
Overall (n=10)§	4.7±0.4	3.9±0.8	0.5±0.3	94±2.2	115±2.0	154±3.5	127±4.0	101±2.9	105±4.9	5.7±1.2	17.1±2.9	19.4±8.0
Mean	4.1±0.21	—	2.2±0.30	134±0.6	—	186±1.2	129±1.5	—	121±2.4	7.0±0.7	—	11.9±2.2
S.e. for variety	1.03	—	0.87	1.94	—	1.88	7.87	—	7.8	4.38	—	14.4

†Sheath brown rot disease incidence (%) at maturity; ‡R = red grain, W = white grain and Chhomrong Dhan, Darmali, and Jumli Marshi are standard local control varieties; §Mean of five best yielding entries and all ten entries from conventional breeding, Nepal Agricultural Research Council (NARC) selection from IRRI nurseries and NARC breeding materials.

produced cultivars which are performing well in the conventional trials system and which are spreading in farmers' fields. In a conventional breeding system, material such as M-3G and M-3C would still be at a preliminary stage of varietal screening in very small plots. It would be at least seven years before they were given to farmers to grow in minikit tests. A release proposal can be submitted three years earlier than in the conventional system, even if time is allowed for selecting for greater uniformity within a farmers' cultivar in order to satisfy seed certification requirements.

The area cultivated with the preferred cultivars increased rapidly in the first three years of the PPB programme. In the conventional system there is a long period, typically five to six years, after release before appreciable adoption occurs (Morris *et al.*, 1992). Some farmers have already started distributing seed of the preferred varieties to other farmers, and it seems likely that M-3G will be cultivated widely in domains similar to Ghandrak and Chhomrong by the time the variety is released through the formal system.

A large number of exotic, cold-tolerant rice varieties have been supplied from the International Rice Research Institute through the International Rice Cold Tolerance Nursery and many have been included in the National Rice Cold Tolerance Nursery. These nurseries were evaluated at Chhomrong, and the majority of the entries failed to set grain in villages at high altitudes (Sthapit, 1992). In comparison, the varieties developed jointly with farmers were far superior to the best entries from the conventional system (Table 7). The advantage of using locally adapted parents and selection in the target environment can be seen over the introduction of international nurseries. A farmer-selected variety will be acceptable to the farmer but it may also yield well in the formal varietal testing system. The ability of M-3 to perform well in both systems was partly because trial locations were situated so as to represent farmers' non-problem fields, and also because the trial management did not differ radically from that used in farmers' fields. Furthermore, M-3 is adapted to both stressed and non-stressed environments.

There were significant differences between farmers' and researchers' strategies for selecting test sites. It is a common practice in formal research systems to use good, uniform land for trials. In contrast, farmers avoid risk by testing new materials first on their worst land, where severe abiotic stresses occur, and then growing them on better fields if they have performed well. This strategy shows how difficult it is to represent the heterogeneous environments of farmers' fields in uniform on-station conditions. Any cultivar from the centralized system will first have to pass the test of being grown in farmers' poorest fields, a factor which is only taken into account fully in PPB. The farmers' risk aversion strategy shows that they are not exposed to unjustifiable risk, which is a common criticism levelled at participatory approaches.

Careful choice of farmers is important for the success of PPB. Farmer methods of plant selection varied with the farmers' knowledge and circumstances. M-3 was given to three farmers, but only two of them succeeded in selecting superior types

from it. The farmer who failed to identify superior types had only used mass selected for grain colour. Women farmers were particularly skilful in assessing post-harvest traits such as milling recovery, cooking and eating quality. Men farmers showed more skill in the assessment of standing crops for yield potential, and management requirements.

Farmers evaluate new varieties at all stages of crop growth and the range of traits they consider is far more than breeders can evaluate in trials. Farmers paid particular attention to the crop near to maturity and also at threshing, milling and consumption. Although farmers' involvement in formal variety testing in Nepal is limited to preference ranking at maturity, most of the farmers decided to retain or reject the variety at the post-harvest stage. For example, the area under M-4 expanded because farmers liked its good milling recovery, whereas M-2 was rejected due to its peculiar smell when cooked. M-3 was preferred at all stages because of its yield potential, straw height and good post-harvest traits. Nilgiri-1 was selected by a farmer in Lumle village, but he abandoned his plan to expand the variety in his remaining good fields, because of the high incidence of grain shattering and the poor taste of the variety. Himchuli-2 was rejected by farmers at Chhomrong, Ghandruk and Lumle because rainfall is very high at the time of maturity, causing pre-germination before harvest. In contrast, Himchuli-2 was liked by farmers in Patlekhet village (1500–1700 m), where rainfall and humidity were lower at the time of maturity. In Lumle, the mother of D. Devkota selected non-sprouted panicles and planted the seed from them in a 100-m² area to see whether the problem of sprouting would continue. Such examples strongly support the decentralization of selection.

The study also found that farmers have particular preferences for different cooking and eating quality traits of rice. Soft white grain, which expanded after cooking was preferred. Whether these criteria can be related to standard grain quality testing procedures needs to be investigated. The present varietal testing systems have no mechanisms for considering the varietal traits, such as post-harvest quality, which are important to farmers. Laboratory measurement of grain quality traits is important, but in the formal varietal testing system in Nepal grain quality is often not assessed until a variety reaches the final stage of release (Sthapit, 1995).

As anticipated, PPB increased biodiversity in the participating villages (Witcombe *et al.*, 1996). Varietal diversification has been achieved repeatedly and a dynamic form of genetic diversity will persist as farmers select plants for specific niches. An example is the retention by one farmer of the otherwise unacceptable variety Nilgiri-1 to be grown on one small plot where an inlet of cold stream-water caused high spikelet sterility in local landraces. Cultivars derived from the same F₅ bulks looked different and, even when similar, were likely to contain significantly different latent genetic variation within the population (Berg *et al.*, 1991). PPB has resulted in a dynamic form of *in situ* genetic conservation by farmer participation in the selection of materials generated from landrace × exotic crosses (Witcombe *et al.*, 1996).

The decentralized selection of segregating material from a few carefully chosen crosses, drawing on the active participation of expert farmers, presents an attractive prospect for fostering a more sustainable and productive agriculture for diverse risk-prone environments. The prerequisites of the method are that objectives are clearly identified, local material is involved in the crosses, farmers are willing participants, and breeders are flexible enough to learn from and work with farmers.

The institutionalization of decentralized technology is a challenging issue (Ashby and Sperling, 1994). However, participatory plant breeding (PPB) has already been institutionalized at the Lumle Agricultural Research Centre, after considering the results demonstrated by the work reported in this paper. If the success of these efforts is to be sustained, incentives are required for field staff to work in difficult areas. PPB can deliver the wide range of rice varieties which are needed for farmers' diverse socio-economic circumstances. The conventional approaches of plant breeding which concentrate on grain yield and wide adaptability are not always feasible in developing countries where research resources and trained manpower are limited.

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